Week 12: 16 Nov - 20 Nov 2026

Subject/الموضوع: Environmental science | Unit Title/عنوان الوحدة: Chapter 5 — Population Ecology | Grade/12: الصف

| Lessons الحصص | Learning targets أهداف التعلم | Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز | Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور |
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| Lesson 1<br>5.4 Fertility is Influenced by Culture | I can analyze how cultural, social, and economic factors influence fertility rates by explaining cause-and-effect relationships between family decisions and population change. | Formative assessment<br>• Warm-up: factors influencing family size<br>• MUST: identify pronatalist and antinatalist influences<br>• SHOULD: analyze how education, income, healthcare, and social security affect fertility<br>• COULD: CER evaluating one strategy that may influence fertility responsibly<br>• Exit-ticket success-criteria reflection | McGraw Hill Environmental Science — Chapter 5, Section 5.4<br>Lesson slides and notes available on Teams<br>Class activities / follow-up available through Toddle |
| Lesson 2<br>5.5 The Demographic Transition | I can analyze how birth rates and death rates change through the demographic transition by interpreting stage patterns and explaining their effect on population growth. | Formative assessment<br>• Warm-up: order the demographic-transition stages<br>• MUST: identify the characteristics of each stage<br>• SHOULD: interpret birth/death-rate patterns and explain population change<br>• COULD: CER evaluating a sustainable population-development strategy<br>• Self-check using model answers and exit reflection | McGraw Hill Environmental Science — Chapter 5, Section 5.5<br>Demographic-transition diagrams / lesson materials on Teams<br>Practice tasks and submissions through Toddle |
| Lesson 3<br>Lab: Observing Population Growth with Duckweed | I can analyze population growth experimentally by collecting duckweed population data, graphing change over time, and explaining how environmental conditions may affect growth. | Formative practical investigation<br>• Observe and count duckweed population samples<br>• Record data accurately in a results table<br>• Construct / interpret a population-growth graph<br>• Identify variables and possible limiting factors<br>• Use evidence to explain the observed growth pattern<br>• Lab reflection / teacher observation | Duckweed population-growth lab sheet<br>McGraw Hill Environmental Science — Chapter 5 population concepts<br>Lab safety instructions and class data<br>Graphing support / resources available on Teams |
| Lesson 4<br>Final Project Check & PPT Preparation | I can evaluate the completeness and quality of my final environmental project by checking evidence against the rubric and organizing a clear scientific presentation. | Summative project final checkpoint<br>• Individual workbook / evidence check against rubric<br>• Verify scientific accuracy, sources, map, interaction web, graph, solutions, and action plan<br>• Teacher conferencing and targeted feedback | Term 1 Summative Project workbook and rubric<br>Student research sources and collected evidence<br>PowerPoint / presentation template and Teams resources<br>Toddle submission instructions |

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| | | • Develop / refine final PPT presentation<br>• Final readiness checklist and submission planning | |
| Teacher's notes/ Reminders<br>ملاحظات المعلم والتذكير | Hello Parents,<br>This week students will continue Chapter 5 by studying Section 5.4: Fertility and Culture and Section 5.5: The Demographic Transition. Students will then complete a practical investigation, "Observing Population Growth with Duckweed," where they will collect data, graph population change, and connect their observations to limiting factors and population-growth concepts. In the final lesson, students will complete a final project checkpoint, review their evidence against the rubric, and prepare/refine their project presentation.<br>Please encourage your child to review Chapter 5 lesson materials, bring all required project evidence, and use Teams and Toddle resources to complete outstanding work before final submission.<br>Thank you for your continued support. | | |

Week #12 / 12 الأسبوع

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| **Technology Integration**<br>**دمج التكنولوجيا** | Students use Chapter 5 digital lesson resources, demographic-transition diagrams, shared class/lab data, graphing tools, research sources, and presentation software to analyze population concepts, complete the duckweed investigation, and prepare the final environmental project presentation. |
| **Cross-curricular links**<br>**الروابط بين المواد الدراسية** | Biology: population growth, reproductive strategies, limiting factors, survivorship, and population regulation.<br>Mathematics: tables, graphing population change, interpreting trends, rates, and demographic data.<br>Social Sciences: relationships among culture, education, economics, development, and fertility.<br>English/Communication: CER reasoning, evidence-based explanation, lab conclusions, and scientific presentation skills. |
| **External Assessment Integration دمج التقييمات الخارجية** | Students practice:<br>• interpreting demographic and population-growth evidence;<br>• explaining cause-and-effect relationships using scientific vocabulary;<br>• collecting, organizing, and graphing experimental data;<br>• using evidence to justify conclusions and evaluate strategies;<br>• applying assessment rubrics and success criteria to improve final work. |
| **Wellbeing/ ATL Skills**<br>**الصحة النفسية/ أساليب التعلم** | Critical Thinking: Analyze demographic patterns, experimental evidence, and environmental consequences.<br>Research: Use reliable sources and select evidence relevant to the final project.<br>Communication: Present scientific findings clearly using graphs, CER, and a structured presentation.<br>Self-management: Meet project checkpoints, prepare materials, and respond to teacher feedback.<br>Collaboration: Share class lab observations responsibly while maintaining individual analysis and project ownership. |
| **Standards**<br>**المعايير** | HS-LS2-1 — Use mathematical and/or computational representations to support explanations of factors that affect carrying capacity of ecosystems at different scales. |

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| | HS-LS2-2 — Use mathematical representations to support and revise explanations based on evidence about factors affecting biodiversity and populations in ecosystems.<br>HS-LS2-6 — Evaluate claims, evidence, and reasoning about how complex interactions and changing conditions affect ecosystem stability.<br>HS-ESS3-4 — Evaluate or refine a technological solution that reduces impacts of human activities on natural systems. |
| **Inquiry/ TOK Question**<br>**البحث والاستقصاء أسئلة TOK** | How do biological evidence, cultural context, and population data work together to help us make responsible decisions about population growth and environmental sustainability? |
| Lesson-specific success criteria | Lesson 1 — 5.4 Fertility is Influenced by Culture<br>Learning Target: I can analyze how cultural, social, and economic factors influence fertility rates by explaining cause-and-effect relationships between family decisions and population change.<br>MUST — DOK 1–2: I can identify pronatalist/antinatalist influences and describe factors that affect fertility.<br>SHOULD — DOK 3: I can analyze how education, income, healthcare, social security, and culture influence fertility patterns.<br>COULD — DOK 4: I can evaluate a responsible strategy and justify its likely effect on fertility using CER.<br><br>Lesson 2 — 5.5 The Demographic Transition<br>Learning Target: I can analyze how birth and death rates change through the demographic transition by interpreting stage patterns and explaining their effect on population growth.<br>MUST — DOK 1–2: I can identify the main characteristics of the demographic-transition stages.<br>SHOULD — DOK 3: I can analyze birth/death-rate evidence and explain why population growth changes between stages.<br>COULD — DOK 4: I can evaluate a development strategy and justify how it could support a sustainable demographic transition.<br><br>Lesson 3 — Duckweed Population Growth Lab<br>Learning Target: I can analyze population growth experimentally by collecting duckweed data, graphing change over time, and explaining how environmental conditions may affect growth.<br>MUST — DOK 1–2: I can identify variables, record accurate observations, and describe the population trend.<br>SHOULD — DOK 3: I can analyze the graph/data and explain how one or more limiting factors influence growth.<br>COULD — DOK 4: I can evaluate limitations in the investigation and propose an evidence-based improvement or follow-up investigation.<br><br>Lesson 4 — Final Project Check & PPT Preparation<br>Learning Target: I can evaluate the completeness and quality of my final environmental project by checking evidence against the rubric and organizing a clear scientific presentation.<br>MUST — DOK 1–2: I can verify that all required project evidence and sources are complete and accurate.<br>SHOULD — DOK 3: I can explain how my evidence supports my environmental analysis and proposed response.<br>COULD — DOK 4: I can evaluate weaknesses, trade-offs, and likely outcomes and refine my final action plan and presentation accordingly. |
| **National Identity Domain**<br>**مجالات الربط بالهوية الوطنية** | Belonging<br>• Students connect responsible population and resource decisions to their role as future contributors to UAE sustainability.<br>Global Understanding |

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| | • Students compare demographic patterns and environmental pressures while considering the UAE within a global context.<br>Heritage<br>• Students connect responsible use of water, land, energy, and coastal/desert ecosystems with the UAE's long-standing relationship with its natural environment.<br>National Responsibility<br>• Students apply scientific evidence to propose practical environmental actions that support sustainable development in the UAE. |